Practical 3a: Arithmetic Methods via C# Delegates

AIM : To create arithmetic methods add(), multiply(), subtract(), divide() and call them using C# delegates.

PROGRAM CODE

```
using System;

namespace DelegateArithmetic
{
   delegate double MathOp(double a, double b);

   class Arithmetic
   {
      public static double Add(double a, double b)      => a + b;
      public static double Multiply(double a, double b) => a * b;
      public static double Subtract(double a, double b) => a - b;
      public static double Divide(double a, double b)
      {
         if (b == 0) throw new DivideByZeroException();
         return a / b;
      }
   }

   class Program
   {
      static void Main(string[] args)
      {
         Console.WriteLine("=== Arithmetic via Delegates ===");
         double x = 10, y = 5;

         MathOp op = Arithmetic.Add;
         Console.WriteLine($"Add({x},{y})       = {op(x, y)}");

         op = Arithmetic.Subtract;
         Console.WriteLine($"Subtract({x},{y}) = {op(x, y)}");

         op = Arithmetic.Multiply;
         Console.WriteLine($"Multiply({x},{y}) = {op(x, y)}");
```

```
            op = Arithmetic.Divide;
            Console.WriteLine($"Divide({x},{y})   = {op(x, y)}");

            Console.ReadKey();
        }
    }
}
```

OUTPUT

```
=== Arithmetic via Delegates ===
Add(10,5)      = 15
Subtract(10,5) = 5
Multiply(10,5) = 50
Divide(10,5)   = 2
```